



Assessment of water-quality in the California Sacramento-San Joaquin Delta during a North Delta directed flow action: August- October 2018

Methods

The study is based on the assessment and comparison of broad-area high-resolution water quality mapping surveys of the northern Delta conducted between August - October 2018. The survey consists of four days of high-resolution mapping beginning and ending on the Sacramento River at Rio Vista. Data were collected at high speed to minimize to the extent to which tidal currents obscure nutrient and phytoplankton distributions. Data-collection cruises were conducted under three different environmental/flow conditions in August, September, and October 2018 that coincided with conditions prior to, during, and following the flow action - or pulse - of agricultural return water organized by the California Department of Water Resources and California Natural Resources Agency. Data were collected on each day to target water-quality conditions near slack water following ebb and flood tides.

Definition of regions and reaches for analysis

The study took place across two subregions of the Delta according to Bergamaschi et al. (2020) – including the Western Tidal Zone and Cache Slough Complex. The subregions are further described in Brown et al., (in prep.) but briefly, the Western Delta is considered a tidally forced tidal lagoon and the Cache Slough Complex is a network of terminal channels and were the majority of sample collection occurred.

Sampling and analysis

Underway water sampling and onboard analyses:

The survey was conducted using the USGS R/V Mary Landsteiner or R/V Dora Henry on August 23, 2018, September 5, 2018, September 13, 2018, and October 4, 2018. Data were collected during daylight hours beginning at approximately 7:30 am. For analysis, sample water was continuously pumped onto the boat while underway at speeds up to 30 mph (13 m/s) using a pick-up tube mounted at a fixed depth of approximately 1 m below the surface, routed through a screen to remove large debris and into a pressure-compensated manifold that maintained system pressure at a proscribed level irrespective of boat speed (Downing et al., 2016). A 2-stage debubbler was used to remove bubbles that can interfere with optical measurements. Sample water was split into separate flow paths for each analytical flow path, with constant flow rates maintained by metering the discharge line. Flows were continuously monitored using a sight gauge for each flow path.

The manifold delivered water to the following flowpaths: (1) A flow-through system consisting of a thermosalinograph that recorded temperature and conductance (Sea-Bird Scientific SB45 (TSG), Belluve, WA); fluorometers that measured fluorescence of chlorophyll (fCHL) and fluorescence of dissolved organic matter (fDOM, WETLabs WETstar (WS), Philomath, OR); a beam transmissometer that recorded transmittance and attenuation (WETLabs model C-Star transmissometer (CStar), Philomath, OR), a nitrate (NO₃) analyzer (SUNA V1; Sea-Bird Scientific, Bellevue, WA). (2) A flow chamber for a multiparameter water quality sonde (YSI EXO2; Xylem Inc. (EXO)), Rye Brook, New York) equipped with sensors to measure temperature, specific conductance (SpC), turbidity, pH, dissolved oxygen (DO), fDOM and fCHL.

All instrumentation was cleaned, and calibrations were checked prior to each use following the manufacturer's recommendation or as described below. Data for most instruments was recorded at 1-second frequency on a single data logger (CR6, Campbell Scientific, Logan, Utah, United States) together with a timestamp and boat position obtained from a high-resolution global positioning system receiver (16X-HVS, Garmin, Olathe, Kansas). The FluoroProbe logged data internally and to the host software. All data were displayed in real time, so scientists on board could respond when they observed changes relevant to study objectives, or if issues with flow or instrument performance were noted.

Discrete water sampling and laboratory analyses:

Discrete water samples were collected either underway or—in the case of more extensive sample requirements—while stopped on station. Discrete samples were collected while underway either using a fourth flow path on the sampling manifold or using a separate system with a separate pick-up tube mounted adjacent to the main sampling pick-up and pumping sample through a 0.45-micron filter into a sample bottle. At designated sampling stations (approximately 10 per day) samples for determination of NH₄, NO₃, PO₄, dissolved organic carbon (DOC), and total

dissolved nitrogen (TDN) concentrations were collected using this system. Additionally, unfiltered samples were collected approximately from 1 m depth using a submersible pump for analysis of chlorophyll-a (Chl-a), pheophytin, phycocyanin, allophycocyanin, and phycoerythrin pigments.

All samples were stored on ice in the dark during transit to the laboratory where they were stored at 2°C. Dissolved organic carbon samples were preserved to <2 pH with high-purity sulphuric acid the day following collection and stored at 2°C. Samples for pigment analysis were filtered <24 hours after collection and immediately frozen (-90°C). Total Chl-a, pheophytin, phycocyanin, allophycocyanin, and phycoerythrin pigment analysis samples were filtered through a 0.3 µm nominal pore size glass fiber filter (Advantec MFS, Inc, Dublin, CA). Frozen Chl-a filters and chilled nutrient samples were shipped to the USGS National Water Quality Laboratory (Denver, CO). Frozen filters for analysis of phycocyanin, allophycocyanin, and phycoerythrin were shipped to BSA Environmental Services, Inc. (Beachwood, OH).

Concentrations of nutrients were determined colorimetrically using an automated-segmented flow analyzer. Concentrations of nitrogen as nitrite (NO₂-N) and as nitrate plus nitrite (NO₃-N + NO₂-N) were determined by colorimetric analysis (Fishman, 1993; Patton and Kryskalla, 2011), ammonium as nitrogen (NH₄-N) was determined by colorimetric analysis after reaction with salicylate-hypochlorite (Fishman, 1993), and total dissolved nitrogen (TDN) was determined by alkaline persulfate digestion (Patton and Kryskalla 2003). Orthophosphate as phosphorous (PO₄-P; also referred to as soluble reactive phosphate) was determined by colorimetric analysis after reaction with NH₄ molybdate and reduction with ascorbic acid (Patton and Truitt, 1992). Chlorophyll-a and pheophytin concentrations were determined according to EPA Method 445.0 (Arar and Collins, 1997) using sonication extraction. Dissolved organic carbon concentrations were measured using a total organic carbon analyzer (TOC-VCSH, Shimadzu Scientific Instruments, Columbia, Maryland) using high-temperature catalytic combustion according to a modified version of USEPA method 415.3 (Potter and Wimsatt, 2005). Dissolved organic carbon concentrations were measured using a total organic carbon analyzer (TOC-VCSH, Shimadzu Scientific Instruments, Columbia, Maryland) using high-temperature catalytic combustion according to a modified version of USEPA method 415.3 (Potter and Wimsatt, 2005). Absorbance spectra and fluorescence EEMs were simultaneously measured using an Aqualog® equipped with a photodiode and charge-coupled device (CCD) (Horiba Instruments, NJ, USA) according to the method described by Hansen et al. (2018). Briefly, excitation and absorbance spectra were measured using a 150 W xenon lamp, a 5 nm bandpass, and a 1 s integration time at wavelengths of 240–600 nm. Emission spectra were collected with a CCD at approximately 1.64 nm (4 pixel) intervals at wavelengths of 250–600 nm. To reduce ultraviolet light exposure of the sample and limit the effects of photobleaching, excitation and absorbance wavelengths were scanned from low to high energy (i.e., visible to ultraviolet range) during analysis. Peaks, ratios, and slopes reported in the data were calculated as described by Hansen et al. (2016).

Phycocyanin, allophycocyanin, and phycoerythrin were determined by azolectin-CHAPS buffer extraction as described by (Zimba, 2012) and modified from (Bennett and Bogorad, 1973) at BSA Environmental Services. Picocyanobacteria direct counts were enumerated via epifluorescence microscopy as described by (Murrell and Lores, 2004). Phycoerythrin-containing cells (PE) and phycocyanin-containing cells (PC) were counted separately and then summed to obtain total picocyanobacteria densities for cells sized between 0.2 and 2.0 microns.

Data processing

All data directly logged to the flow-through data collection system was processed using pandas software library (0.220) in Python (3.7.3) to remove periods of compromised data (e.g. flow blockages, bubbles), to apply instrument corrections and unit conversions, and to apply a centered 20 second median filter to the time series. Final NO₃ values were obtained by regressing instrument response against NO₃ concentrations obtained from laboratory measurements of discrete samples collected through the course of the day. Individual sample results with more than three standard deviations from the regression were judged to be outliers and removed from the regression.

Larger differences between SUNA and laboratory NO₃ concentration data are assumed to arise from 1) actual spatial heterogeneity in environmental NO₃ concentrations between water entering the flow through system and that captured in the sample bottle, 2) interferences of turbidity and dissolved organic matter using the UV absorption method for nitrate, or 3) errors in sample handling and laboratory analysis. Excepting these potential sources of error, the field and laboratory NO₃ measurements are typically well correlated with a linear fit. For the simple linear regressions, the SUNA NO₃ concentration data was the explanatory variable to compute the response variable of the laboratory NO₃ concentration. P-values were <0.0001 for all regressions. Summary statistics and regressions for individual days are identified in Table 1.

Dissolved inorganic nitrogen (DIN) concentration was calculated as the sum of NO₃, NO₂ and NH₄ for the discrete sample data. Laboratory DON concentration was calculated by subtracting NH₄, NO₃, and NO₂ data from the TDN data (i.e. $DON = TDN - DIN$). Continuous DON and DOC values were obtained by regressing fDOM sensor (EXO) response against DON and DOC concentrations obtained from laboratory measurements of discrete samples collected across four mapping dates according to modified methods by Snyder et al. (2018). Individual sample results with more than three standard deviations from the regression were judged to be outliers and removed. The regression equation for estimated DON = $0.2769 \cdot (fDOM\ EXO) + 3.817$ with $R^2 = 0.8705$. The regression equation for estimated DOC = $0.0648 \cdot (fDOM\ EXO) + 0.6686$ with $R^2 = 0.9579$.

Data was spatially aligned (SA) to a common spatial framework to facilitate comparisons between dates and to minimize the effects of differential data density on the spatial interpolation calculations by assigning median-filtered data by proximity to a centerline vector comprising points located at the centerline of all channels navigated, spaced at approximately 150 m intervals. The raster output was smoothed with the Focal Statistics tool, and interpolated values were extracted to the spatially aligned points to create continuous spatially aligned data across the mapping domain.

Table 1. Summary statistics and linear regression for nitrate (NO₃) concentration data obtained from high-resolution SUNA instrument and discrete samples analyzed using standard laboratory methods [n, number of observations; μ M, micromolar; mean, mean high-resolution nitrate concentration; median, median high-resolution nitrate concentration; min, minimum high-resolution nitrate concentration; max, maximum high-resolution nitrate concentration; SS, sum of squared error; SSE, sum of squared estimate of error; MSE, mean square error; RMSE, root mean square error; %, percent; MSPE, model standard percentage error; R², coefficient of determination; *, asterisk indicates adjusted R²].

Parameter	n	min (μ M)	max (μ M)	median (μ M)	mean (μ M)	SS (μ M)	SSE (μ M)	MSE (μ M)	RMSE (μ M)	MSPE (%)	regression equation	p-value	R ²
NO ₃ (SUNA)	29	1.8	12.4	6.9	5.9	269	9.7	0.4	1.0	16.8	y = 1.3683x - 2.4418	<0.0001	0.9265*

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